

Latest WIRELESS STANDARDS And Their APPLICATIONS

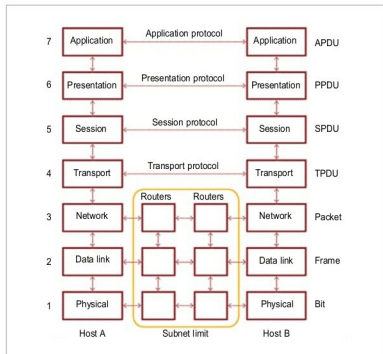
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With newer technologies like 5G and IoT, standardisation becomes necessary for the devices to communicate with each other

The cool quotient of communication devices is increasing immensely as we advance towards newer wireless means for connectivity. The need to update benchmarks for these systems also arises alongside. The latest wireless communication technology standards have to adhere to the requirements and ensure better security and quality than the previously adopted standards.

According to Bipin Pradeep Kumar, co-founder and director, Gaia Smart Cities, "Standardisation is a complex subject that needs to be looked at different levels. At the communication level, apart from Wi-Fi, unlicensed long-range technologies like Sigfox and LoRaWAN are being adopted widely. Then there are several short-range technologies as well, like Zigbee and z-wave. All these technologies should be able to coexist. On the protocol side, there are so many options. Prominent among them is MQTT (Message Queue Telemetry Transport), which is an ISO20922 standard."

Fig. 1: Open Systems Interconnection (OSI) reference model (Image source: <http://media.techtarget.com>)



Device-to-device communication standard

Direct device-to-device (D2D) communications technology is being considered to provide low-power, high-data-rate and low-latency services to 5G network users.

Rajan Mathews, director general, Cellular Operators Associations of India (COAI), says, "The latest iteration of mobile broadband technology, 5G, will bring in new unique network and service capabilities." "It will be a thousand times faster than 4G and will provide Internet access in areas that have not yet been tapped because of various roadblocks," he adds.

5G is hopping to a new level where innumerable devices can be connected to each other using D2D standard and where security breach is extremely difficult. This D2D technology would be beneficial even when the network is out of range. A wide array of sensor networks can be used for home security systems, energy grids and entertainment.

Basically, 5G is 4G technology integrated with worldwide wireless web. Currently, in terms of bandwidth, it offers more than 1Gbps as against 4G's 200Mbps.

"We are very excited about the possibilities this opens up in terms of network architecture, particularly in view of 5G's aim to connect new industries and empower unique service scenarios," said Woojune Kim, vice president and head of next-generation strategy, Samsung Electronics, in a press release.

D2D standard was introduced by Third Generation Partnership Project (3GPP)—one of the most powerful standardisation bodies in the world. 3GPP brings together seven different telecom development associations, namely, Association of Radio Industries and Businesses (ARIB), Alliance for Telecommunications Industry Solutions (ATIS), China Communications Standards Association (CCSA), European Telecommunications Standards Institute (ETSI), Telecommunications Standards Development Society, India (TSDSI), Telecommunications Technology